	Using Fleming's left hand rule, since magnetic force on wire is upwards and current is from Y to X, the magnetic field must be from A (North) to the opposite face (South)
22.	**Ans: A** As solenoid X moves away from Y, Y will experience a decreasing magnetic flux through it, therefore, the direction of the induced e.m.f. will be so as to create a magnetic field in the same direction to oppose the decrease. By right hand grip rule, the current will flow from N to M. Since the magnetic North pole for Y is facing the magnetic South pole for X, the force will be attractive in nature. This can also be explained by Lenz's law as the solenoid Y will need to be attracted so that the motion of solenoid X away from it can be opposed. Lastly, the rate of change of magnetic flux linkage will be decreasing as the magnetic flux density will not be dropping at a linear rate but follows an inverse cube relationship. Therefore current is diminishing.
23.	**Ans: A**